

Sequence defined on multisets

OPG-37226	L1 - Tractable	Partiellement résolu	Combinatorics
-----------	----------------	----------------------	---------------

Énoncé original (anglais, reproduction mot pour mot)

Conjecture Define a $2 \times n$ array of positive integers where the first row consists of some distinct positive integers arranged in increasing order, and the second row consists of any positive integers in any order. Create a new array where the first row consists of all the integers that occur in the first array, arranged in increasing order, and the second row consists of their multiplicities. Repeat the process. For example, starting with the array $[1; 1]$, the sequence is: $[1; 1] \rightarrow [1; 2] \rightarrow [1, 2; 1, 1] \rightarrow [1, 2; 3, 1] \rightarrow [1, 2, 3; 2, 1, 1] \rightarrow [1, 2, 3; 3, 2, 1] \rightarrow [1, 2, 3; 2, 2, 2] \rightarrow [1, 2, 3; 1, 4, 1] \rightarrow [1, 2, 3, 4; 3, 1, 1, 1] \rightarrow [1, 2, 3, 4; 4, 1, 2, 1] \rightarrow [1, 2, 3, 4; 3, 2, 1, 2] \rightarrow [1, 2, 3, 4; 2, 3, 2, 1]$, and we now have a fixed point (loop of one array).

The process always results in a loop of 1, 2, or 3 arrays.

Source et attribution. Énoncé reproduit depuis UnsolvedMath, jeu de données officiel sous licence CC BY 4.0.

Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Jeu de données: <https://huggingface.co/datasets/ulamai/UnsolvedMath>